



Lab4 Intro

Writing Basic Software Application

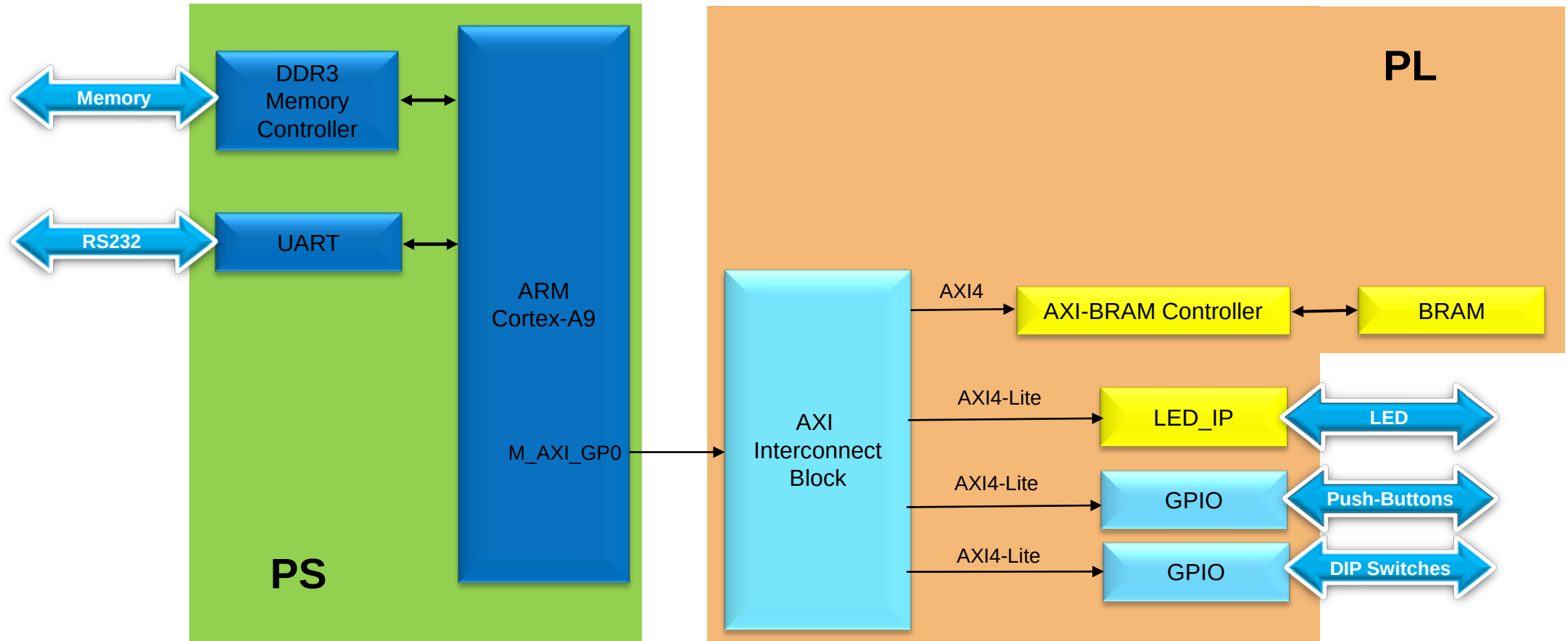
Zynq
Vivado 2014.2 Version

Introduction

- This lab guides you through the process of writing a basic software application. You will add AXI BRAM controller and a BRAM memory.
- Then you will create a software project in SDK and develop a software that will monitor dip switches and write to the LED_IP device to which LEDs are connected.
- You will also modify a linker script file and view its effect on the sections location.
- Finally, you will download the bit file and verify the functionality

ARM Cortex-A9 based Embedded System Design

Lab4: Add BRAM and develop software



Procedure

- **Open project in Vivado**
- **Add an internal BRAM**
- **Generate bitstream and invoke SDK**
- **Create a software project**
- **Analyze assembled object files**
- **Verify the design in hardware**

Summary

- **Use SDK to define, develop, and integrate the software components of the embedded system.**
- **You can define a device driver interface for each of the peripherals and the processor.**
- **SDK imports an xml file and lets you update the settings so you can develop the software side of the processor system.**
- **You can then develop and compile peripheral-specific functional software and generate the executable file from the compiled object codes and libraries. If needed, you can also use a linker script to target various segments in various memories.**